Regularised GLM for Credit Default Prediction

VAE-Augmented Feature Strategies – Implementation & Results

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1 Overview

This document describes the implementation, configuration, and results of the regularised Generalised Linear Model (GLM) pipeline for predicting firm-level credit default. The model uses elastic-net penalisation via `glmnet` and is evaluated against four feature strategies constructed from raw financial data and Variational Autoencoder (VAE) outputs.

The pipeline is self-contained in `04A_Train_GLM.R` and produces all results from disk – no other pipeline stage needs to be active in memory.

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2 Data & Pipeline Position

2.1 Input Files

The GLM reads six files produced by earlier pipeline stages:

```
02_Pipeline Datasets and Results/
|-- 02_Data/
|   |-- 02_train_final_OoS.rds      161,404 rows x 509 cols
|   |-- 02_test_final_OoS.rds        69,017 rows x 509 cols
|   |-- 02_train_id_vec_OoS.rds      Firm IDs for train set
|   `-- 02_test_id_vec_OoS.rds       Firm IDs for test set
`-- 03_Output/Latent/
    |-- latent_train_OoS.parquet     VAE latent dims (z1-z32)
    |-- latent_test_OoS.parquet
    |-- anomaly_train_OoS.parquet    VAE reconstruction error
    `-- anomaly_test_OoS.parquet
```

2.2 Dataset Characteristics

Property	Value
Training observations	161,404
Test observations	69,017
Base features	509
Default rate	~0.8%
Split mode	Out-of-Sample (OoS), stratified 70/30 at firm level

The 70/30 split is performed at **firm level** to prevent data leakage – all observations of a given firm appear entirely in either Train or Test, never both.

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3 Feature Strategies

Four distinct feature sets are constructed and trained independently.

3.1 Base Model – Raw Features

The baseline uses the 509-column feature matrix from `02_FeatureEngineering.R`, including:

- Raw financial ratios (`r1`–`r20`) and balance sheet positions (`f1`–`f11`)
- Time-series dynamics: year-over-year changes, acceleration, momentum, consecutive decline streaks
- Sector deviation features: sector rank, volatility, trend, divergence
- One-hot encoded sector and size dummies

The Base Model serves as the performance benchmark against which all strategies are measured.

3.2 Strategy A – Latent Features Only

Replaces all 509 raw features with the **32 latent dimensions** (`z1`–`z32`) produced by the VAE encoder. These represent a compressed, non-linear embedding of the firm's financial position. The hypothesis is that this compression captures structural patterns not visible in individual ratios.

Feature matrix: 32 columns + target `y`

3.3 Strategy B – Anomaly Score Only

Uses only the VAE **reconstruction error** as a single predictor. A high reconstruction error means the VAE cannot represent the firm's balance sheet from the learned population norm. The hypothesis is that financial anomalousness is itself a default signal.

Feature matrix: 1 column (`vae_recon_error`) + target `y`

3.4 Strategy C – Raw + Latent + Anomaly

Combines all information sources: the 509 raw features, the 32 latent dimensions, and the reconstruction error. This tests whether VAE-derived features add incremental discriminatory power over raw features when combined.

Feature matrix: 542 columns + target `y`

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4 Model Implementation

4.1 Algorithm

Elastic-net penalised logistic regression via the `glmnet` package:

$$\text{Loss} = -\ell(\beta) + \lambda \left[\alpha \|\beta\|_1 + \frac{1-\alpha}{2} \|\beta\|_2^2 \right]$$

where $\ell(\beta)$ is the binomial log-likelihood, λ controls overall regularisation strength, and $\alpha \in [0, 1]$ controls the mix between Lasso ($\alpha = 1$) and Ridge ($\alpha = 0$).

4.2 Hyperparameter Optimisation

The optimal (α, λ) pair is selected via `cv.glmnet`:

Parameter	Value	Rationale
`alpha_grid`	{0, 0.5, 1}	Ridge, Elastic-net, Lasso
`nlambda`	20	Reduced from default 100 for speed
`type.measure`	`"auc"`	Optimise AUC directly
`foldid`	Firm-level 5-fold CV	Consistent across strategies
`standardize`	`FALSE`	Features already processed upstream

4.3 Cross-Validation Design

Folds are assigned at **firm level**, stratified by `y_ever` (whether the firm ever defaults). This ensures no firm appears in both training and validation within any fold, and the fold structure is identical across all strategies.

4.4 Checkpointing

Each trained model is saved as an `.rds` file in `03_Output/GLM/`. On subsequent runs the checkpoint is loaded instead of retraining.

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5 Key Variables

5.1 Configuration

Variable	Value	Description
`SPLIT_MODE`	`"OoS"`	Out-of-Sample split
`SEED`	123	Reproducibility seed
`TARGET_COL`	`"y"`	Binary default indicator
`N_FOLDS_GLM`	5	Number of CV folds
`TOP_N_COEF`	20	Top coefficients reported
`alpha_grid`	c(0, 0.5, 1)	Alpha values in HPO
`nlambda`	20	Lambda path length per alpha

5.2 Output Objects

Variable	Description
`GLM_Results_BaseModel`	Trained model for Base Model
`GLM_Results_Strategy_A/B/C`	Trained models per strategy
`GLM_Leaderboard_Train`	CV AUC leaderboard
`GLM_Leaderboard_Test`	Test AUC leaderboard
`GLM_Metrics`	Full evaluation metric table
`GLM_DeLong`	DeLong pairwise AUC test results

Variable	Description
`predictions_combined`	Stacked predictions all strategies

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6 Results

6.1 CV Leaderboard (Training)

Model	CV AUC	Uplift vs Base	Alpha	Lambda
Strategy C	0.9092	+0.10%	1.00	0.0001
Base Model	**0.9083**	**0.00%**	**1.00**	**0.0001**
Strategy A	0.8304	-8.60%	0.25	0.0001
Strategy B	0.5193	-42.84%	–	–

6.2 Test Set Leaderboard

Model	Test AUC	Uplift vs Base	Brier	Alpha	Lambda
Strategy C	0.9089	+0.06%	0.00770	1.00	0.0001
Base Model	**0.9084**	**0.00%**	**0.00769**	**1.00**	**0.0001**
Strategy A	0.8292	-8.72%	0.00809	0.25	0.0001
Strategy B	0.5628	-38.05%	0.00832	–	–

6.3 Strategy Comparison Chart

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7 Evaluation Metrics

7.1 Full Metric Table

Model	AUC	AP	Brier	BSS	R@FPR10	Sens.	Spec.	F1
M1 Base	**0.9084**	**0.1268**	**0.00769**	**0.0674**	**0.756**	**0.820**	**0.858**	**0.088**
M2 Latent	0.8292	0.0452	0.00809	0.0191	0.502	0.799	0.724	0.046
M3 Anomaly	0.5628	0.0069	0.00832	-0.008	0.109	0.393	0.476	0.012
M4 Combined	**0.9089**	**0.1257**	**0.00770**	**0.0664**	**0.744**	**0.841**	**0.830**	**0.076**

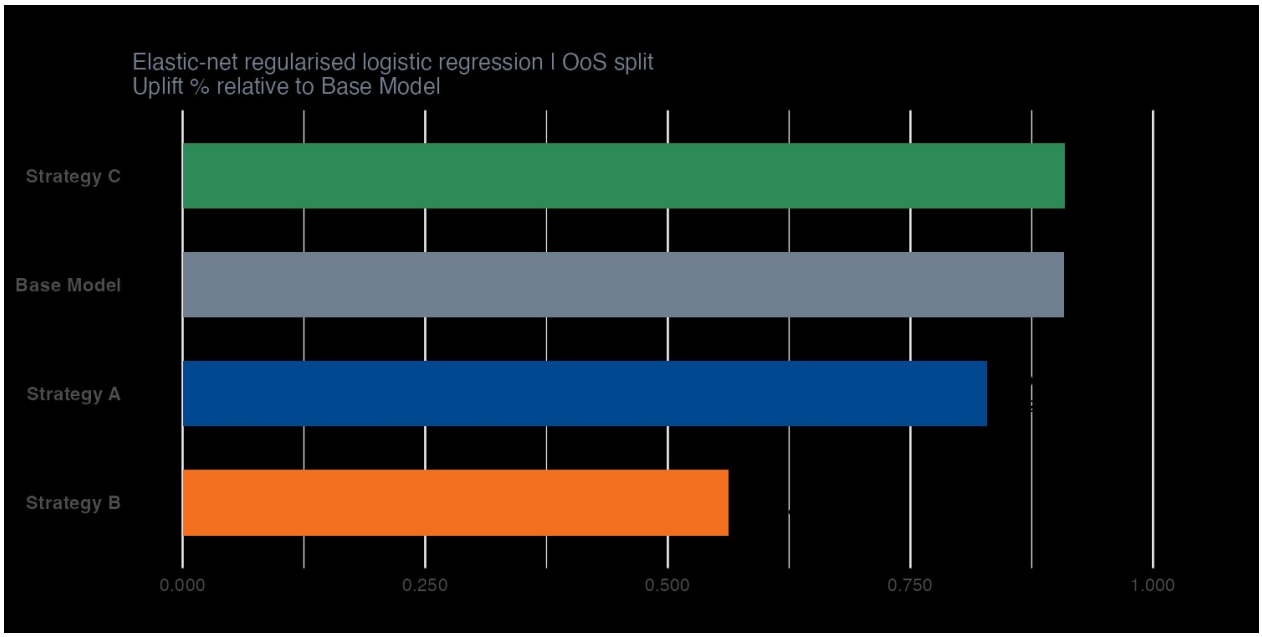

Figure 1: Test Set AUC by Strategy

7.2 Metric Definitions

AUC-ROC – Area Under the ROC curve. Range [0.5, 1.0]; higher is better.

AP (Average Precision) – Area under the Precision-Recall curve. More informative than AUC under severe class imbalance.

Brier Score – Mean squared error of predicted probabilities: $\frac{1}{n}\sum(p_i - y_i)^2$. Lower is better.

BSS (Brier Skill Score) – Improvement over naive forecast: $\text{BSS} = 1 - \frac{\text{BS}}{\text{BS}_{\text{ref}}}$. Positive = better than naive.

R@FPR10 – Recall at 10% False Positive Rate. If a bank flags the 10% of firms it considers most risky, what fraction of actual defaults does it catch? The most practically relevant metric for credit risk management.

Youden Threshold – The probability cutoff that maximises Sensitivity + Specificity − 1.

F1 – Low (~0.09) due to extreme class imbalance (0.8% defaults). Expected.

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8 Statistical Comparison – DeLong Test

The DeLong test performs a non-parametric pairwise comparison of AUC values, accounting for the correlation of paired predictions on the same test set.

Model vs Base	AUC	Base AUC	Delta AUC	95% CI	p-value
M2 Latent-Features	0.8292	0.9084	-0.0792	[-0.093, -0.066]	< 0.001
M3 AnomalyScore	0.5628	0.9084	-0.3456	[-0.371, -0.321]	< 0.001
M4 RawLatentAnomaly	0.9089	0.9084	+0.0005	[-0.001, +0.002]	0.434

Strategy A ($p < 0.001$): Significantly worse than Base. Latent dimensions lose the discriminatory signal in raw features; compression is suboptimal for classification.

Strategy B ($p < 0.001$): Significantly worse than Base. Reconstruction error captures financial unusualness but not the direction of distress.

Strategy C ($p = 0.434$): Not significantly different from Base. The +0.05% AUC improvement is within statistical noise. This is the central finding: VAE features do not meaningfully improve a well-specified GLM.

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9 Key Findings

1. The Base Model is strong. AUC of 0.908 with raw financial ratios and time-series features is excellent. The feature engineering in `02_FeatureEngineering.R` already captures most of the available signal.

2. VAE features do not improve the GLM. Strategy C shows +0.05% AUC ($p = 0.434$) – statistically indistinguishable from zero.

3. Raw features dominate latent representations. Strategy A achieves AUC = 0.829 vs 0.908 for Base – a significant degradation. The VAE's compression, optimised for reconstruction, is suboptimal for default prediction.

These findings are consistent with the credit risk literature: linear models with well-engineered accounting ratios are difficult to improve upon with unsupervised representation learning.

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10 Output Files

File	Content
`GLM_Leaderboard_v2_OoS.xlsx`	AUC, Brier, Uplift – train CV and test
`GLM_Variable_Importance_v2_OoS.xlsx`	Top 20 coefficients per strategy
`GLM_Leaderboard_Chart_v2_OoS.png`	Bar chart of test AUC by strategy
`predictions_test_GLM_v2_OoS.parquet`	`id, y, p_default, model, framework`
`GLM_Evaluation_v2_OoS.xlsx`	Full metrics + DeLong tests

File	Content
`GLM_checkpoint_*.rds`	Saved model objects for fast reloading

The `.parquet` prediction file matches the AutoGluon and XGBoost format and can be stacked directly for cross-framework comparison in `06_Evaluation.R`.

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11 Feature Importance

Feature importance is reported in two complementary ways for each strategy:

- **Train coefficients** – the elastic-net coefficient β_j assigned to each feature by the final model. Larger $|\beta_j|$ means the model places more weight on that feature. The sign indicates direction of risk.
- **Test contributions** – the mean absolute contribution $|x_j \cdot \beta_j|$ across all test observations. This shows which features actually *moved* the predicted probability in practice, accounting for both the coefficient magnitude and the typical scale of the feature values.

11.1 Base Model (M1) – Top Features

The Base Model relies on a mix of sector-relative, temporal, and ratio features.

11.1.1 Top 10 by Coefficient (Train)

Rank	Feature	Coefficient	Odds Ratio	Direction
1	`secVol_r10`	+13.22	549,937	Risk+
2	`history_length`	-1.71	0.18	Risk-
3	`time_index`	+1.39	4.02	Risk+
4	`expmean_r12`	-1.29	0.28	Risk-
5	`sector_retail`	-0.92	0.40	Risk-
6	`expvol_r12`	-0.78	0.46	Risk-
7	`secRank_r7`	+0.72	2.05	Risk+
8	`secRank_f11`	+0.70	2.01	Risk+
9	`expmean_r11`	-0.62	0.54	Risk-
10	`sector_construction`	-0.43	0.65	Risk-

11.1.2 Top 10 by Test Contribution (Mean |x * beta|)

Rank	Feature	Coef	Mean	x*beta
1	`history_length`	-1.71	6.672	-6.672
2	`secVol_r10`	+13.22	3.718	+3.718
3	`time_index`	+1.39	3.403	+3.403
4	`secRank_r7`	+0.72	0.360	+0.360
5	`secRank_f11`	+0.70	0.349	+0.349

Rank	Feature	Coef	Mean	x*beta
6	`expmean_r12`	-1.29	0.324	-0.324
7	`secVol_f2`	-0.36	0.314	-0.048
8	`r7` (Debt Ratio)	+0.37	0.291	+0.001
9	`f8` (Net Profit)	-0.23	0.183	+0.002
10	`sector_retail`	-0.92	0.099	-0.099

11.1.3 Economic Interpretation

The most important drivers align with credit risk theory:

- **`history_length`** (negative) – firms with longer observation history are less likely to default. Young firms with short histories carry higher uncertainty.
- **`secVol_r10`** (positive) – high volatility of the short-term asset structure relative to sector peers signals instability and elevated default risk.
- **`time_index`** (positive) – captures temporal trends; later periods in the panel have higher default probability, consistent with macroeconomic cycles.
- **`secRank_r7`** (positive) – a firm with a high debt ratio relative to its sector peers is at greater risk. Sector-relative ranking is more informative than the absolute ratio.
- **`secRank_f11`** (positive) – high retained earnings rank within the sector is positively associated with default risk; this likely captures firms that have been retaining earnings due to inability to distribute them.
- **`expmean_r12`** (negative) – high exponential moving average of the cash-to- current-assets ratio reduces default risk. Liquidity protects firms.
- **`r7`** (positive) – Debt Ratio; higher leverage increases default probability.
- **`f8`** (negative) – Net Profit; more profitable firms are less likely to default.

11.2 Strategy A (M2) – Latent Features Only

With only 32 latent dimensions, the model assigns interpretable weights to the VAE's compressed representation of the balance sheet.

11.2.1 Top 10 by Coefficient (Train)

Rank	Feature	Coefficient	Odds Ratio
1	`z3`	+1.34	3.82
2	`z26`	+0.97	2.65
3	`z20`	+0.82	2.28
4	`z17`	+0.72	2.05
5	`z16`	-0.69	0.50
6	`z31`	+0.57	1.77
7	`z9`	-0.51	0.60
8	`z8`	-0.49	0.61
9	`z1`	-0.45	0.64
10	`z15`	+0.43	1.54

11.2.2 Top 5 by Test Contribution

Rank	Feature	Mean
1	`z26`	0.791
2	`z3`	0.443
3	`z8`	0.440
4	`z16`	0.426
5	`z15`	0.364

The latent dimensions have no direct economic label – they represent learned combinations of all financial features. The relatively high weight on `z26` and `z3` suggests the VAE has encoded specific balance-sheet configurations that correlate with default, but the information is spread across all 32 dimensions rather than concentrated in a few interpretable features.

11.3 Strategy B (M3) – Anomaly Score Only

Strategy B has a single feature, so importance analysis reduces to one row:

Feature	Coefficient	Odds Ratio	Mean
`vae_recon_error`	+0.0179	1.018	0.012

The positive coefficient confirms the hypothesis – higher reconstruction error (more anomalous firms) is associated with higher default probability. However, the coefficient is very small (+0.018) and the test contribution (0.012) is negligible compared to Base Model features (top feature: 6.67). This explains why Strategy B achieves only AUC = 0.563 – a single scalar anomaly score lacks the directional information needed for accurate default classification.

11.4 Strategy C (M4) – Raw + Latent + Anomaly

With 542 features, the model largely replicates the Base Model's top drivers, with VAE latent dimensions appearing lower in the ranking.

11.4.1 Top 10 by Test Contribution

Rank	Feature	Type	Coef	Mean
1	`history_length`	Raw	-1.77	6.909
2	`time_index`	Raw	+1.44	3.529
3	`secVol_r10`	Raw	+10.70	3.011
4	`secRank_f11`	Raw	+0.93	0.462
5	`secRank_r7`	Raw	+0.86	0.433
6	`secVol_f2`	Raw	-0.38	0.326
7	`r7` (Debt Ratio)	Raw	+0.34	0.272

Rank	Feature	Type	Coef	Mean
8	`has_history`	Raw	+0.36	0.244
9	`expmean_r12`	Raw	-0.88	0.220
10	`f8` (Net Profit)	Raw	-0.27	0.213
11	`z4`	**VAE**	+0.24	0.173
12	`z5`	**VAE**	-0.45	0.163

The first VAE latent dimension (`z4`) appears only at rank 11 in test contributions. This confirms the DeLong test result ($p = 0.434$): the VAE features are selected by the elastic-net but contribute minimally to the final predictions. Raw financial features dominate.

11.5 Cross-Strategy Comparison

The table below shows which features appear in the top 10 across strategies, demonstrating consistency of economic signal:

Feature	Base	Strategy C	Economic Meaning
`history_length`	#1 (test)	#1 (test)	Firm age / data availability
`secVol_r10`	#2 (test)	#3 (test)	Sector-relative short-term asset volatility
`time_index`	#3 (test)	#2 (test)	Temporal trend in default rates
`secRank_r7`	#4 (test)	#5 (test)	Debt ratio rank within sector
`secRank_f11`	#5 (test)	#4 (test)	Retained earnings rank within sector
`r7` (Debt Ratio)	#8 (test)	#7 (test)	Absolute leverage
`f8` (Net Profit)	#10 (train)	#10 (test)	Profitability

The high consistency across Base Model and Strategy C confirms that the elastic- net correctly identifies the same underlying economic drivers regardless of the feature set. The VAE features are not replacing these drivers – they are simply additional, lower-ranked inputs.